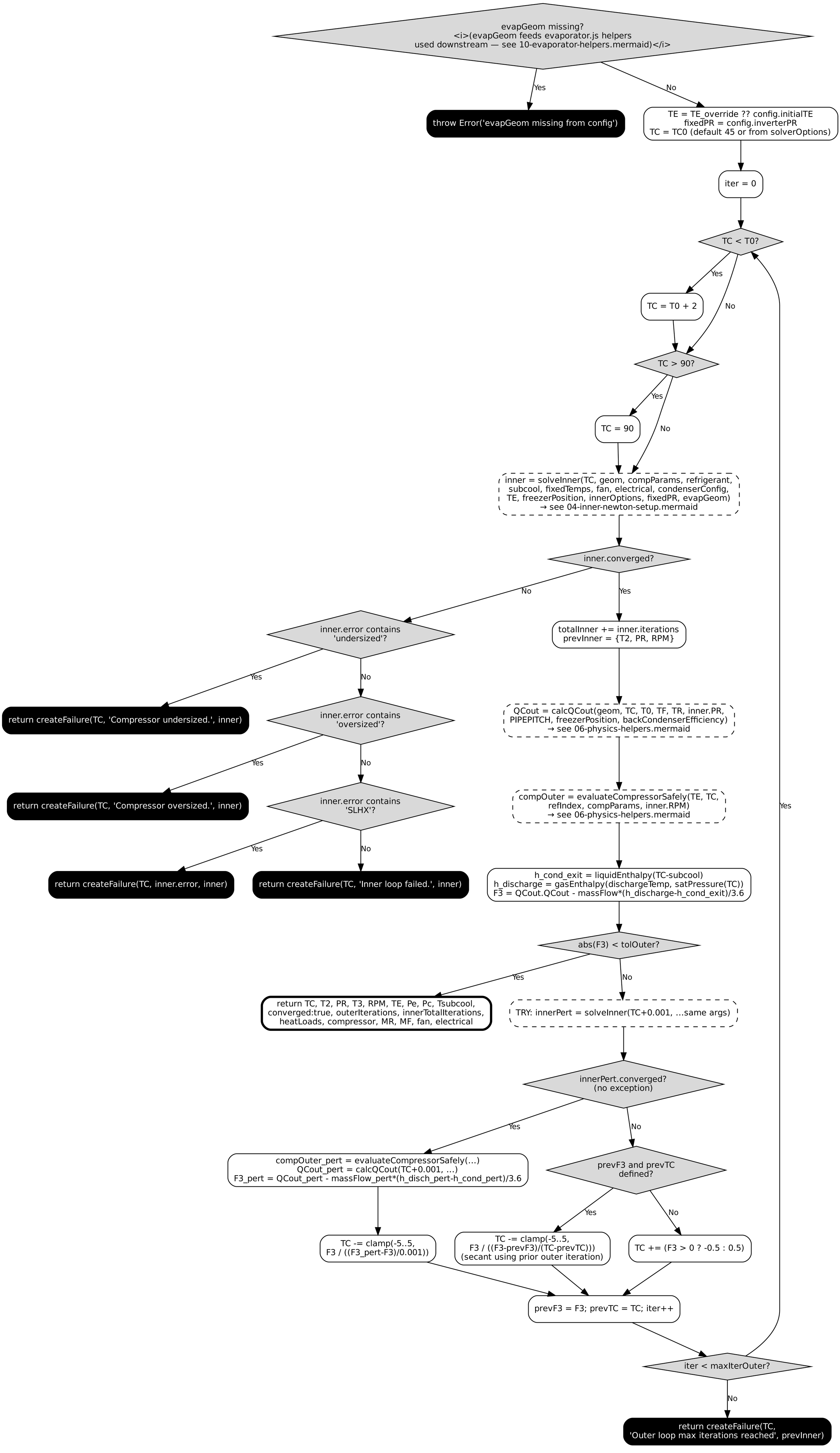
evapGeom missing?
(evapGeom feeds evaporator.js helpers
used downstream — see 10-evaporator-helpers.mermaid)
Yes
No
throw Error('evapGeom missing from config')
TE = TE_override ?? config.initialTE
fixedPR = config.inverterPR
TC = TC0 (default 45 or from solverOptions)
iter = 0
TC < T0?
Yes
No
TC = T0 + 2
TC > 90?
Yes
No
TC = 90
inner = solveInner(TC, geom, compParams, refrigerant,
subcool, fixedTemps, fan, electrical, condenserConfig,
TE, freezerPosition, innerOptions, fixedPR, evapGeom)
→ see 04-inner-newton-setup.mermaid
inner.converged?
No
Yes
inner.error contains
'undersized'?
Yes
No
return createFailure(TC, 'Compressor undersized.', inner)
inner.error contains
'oversized'?
Yes
No
return createFailure(TC, 'Compressor oversized.', inner)
inner.error contains
'SLHX'?
Yes
No
return createFailure(TC, inner.error, inner)
return createFailure(TC, 'Inner loop failed.', inner)
totalInner += inner.iterations
prevInner = {T2, PR, RPM}
QCout = calcQCout(geom, TC, T0, TF, TR, inner.PR,
PIPEPITCH, freezerPosition, backCondenserEfficiency)
→ see 06-physics-helpers.mermaid
compOuter = evaluateCompressorSafely(TE, TC,
refIndex, compParams, inner.RPM)
→ see 06-physics-helpers.mermaid
Yes
h_cond_exit = liquidEnthalpy(TC-subcool)
h_discharge = gasEnthalpy(dischargeTemp, satPressure(TC))
F3 = QCout.QCout - massFlow*(h_discharge-h_cond_exit)/3.6
abs(F3) < tolOuter?
Yes
No
return TC, T2, PR, T3, RPM, TE, Pe, Pc, Tsubcool,
converged:true, outerIterations, innerTotalIterations,
heatLoads, compressor, MR, MF, fan, electrical
TRY: innerPert = solveInner(TC+0.001, ...same args)
innerPert.converged?
(no exception)
Yes
No
compOuter_pert = evaluateCompressorSafely(…)
QCout_pert = calcQCout(TC+0.001, …)
F3_pert = QCout_pert - massFlow_pert*(h_disch_pert-h_cond_pert)/3.6
prevF3 and prevTC
defined?
Yes
No
TC -= clamp(-5..5,
F3 / ((F3_pert-F3)/0.001))
TC -= clamp(-5..5,
F3 / ((F3-prevF3)/(TC-prevTC)))
(secant using prior outer iteration)
TC += (F3 > 0 ? -0.5 : 0.5)
prevF3 = F3; prevTC = TC; iter++
iter < maxIterOuter?
No
return createFailure(TC,
'Outer loop max iterations reached', prevInner)